A Variance-Scaled Confidence-Overlap Strategy for Multi-Armed Bandits

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1 Introduction and Design Motivation

This assignment considers a stochastic multi-armed bandit problem. At each round $t = 1, \dots, T$, the learner selects an arm $a_t \in \{1, \dots, K\}$ and observes a random reward r_t. Each arm i has an unknown reward distribution with mean μ_i. The objective is to minimize cumulative pseudo-regret,

$$R(T) = \sum_{t=1}^{T} (\mu^* - \mu_{a_t}), \qquad \mu^* = \max_i \mu_i.$$

Since the regret is computed from the true means rather than realized noisy rewards, a good algorithm should not simply chase arms that receive lucky early samples. It should estimate expected rewards reliably and concentrate on the best arm as quickly as possible.

The main difficulty is the exploration–exploitation tradeoff. Greedy selection can commit too early to a wrong arm, while ϵ-greedy continues random exploration even after enough information has been collected. Softmax selection is smoother, but its behavior depends heavily on the temperature and does not directly account for the uncertainty of each empirical mean.

While testing the public environments, I observed several different failure modes. In `env1`, the best arm is clearly separated, so fast exploitation is important. In `env2`, the gaps are small, so premature commitment is risky. In `env3`, the best arm has high variance, so it may look bad after unlucky early samples. In `env4`, a high-variance suboptimal arm can look deceptively good. In `env5`, many arms make unnecessary exploration costly.

These observations motivated my final method, the *Variance-Scaled Confidence-Overlap Strategy* (VSCO). The core idea of VSCO is to explore only arms that are still plausible competitors to the current empirical best arm. VSCO uses empirical means for exploitation, an uncertainty bonus for exploration, empirical variance to adjust the bonus size, and a confidence-overlap rule to prune arms that are unlikely to be optimal. The project page is available at: `https://hoonably.github.io/cse-archive/vsco`

2 Proposed Method

For each arm i, VSCO maintains the number of pulls $N_i(t)$, the reward sum $S_i(t) = \sum_{s<t: a_s = i} r_s$, and the squared reward sum $Q_i(t) = \sum_{s<t: a_s = i} r_s^2$. From these values, it computes

$$\hat{\mu}_i(t) = \frac{S_i(t)}{N_i(t)}, \qquad \hat{\sigma}_i(t) = \sqrt{\max\left(0, \frac{Q_i(t)}{N_i(t)} - \hat{\mu}_i(t)^2\right)}.$$

The algorithm begins with a short warm-up phase, where each arm is pulled

$$m_T = \min\left(4, \max\left(1, \left\lfloor \frac{T}{20K} \right\rfloor\right)\right)$$

times. This guarantees initial observations for all arms while preventing excessive forced exploration when T is small.

After warm-up, each arm receives the variance-scaled confidence bonus

$$B_i(t) = \left(0.12 \cdot \frac{T}{T+25}\right) \sqrt{\frac{\log(t+1)}{N_i(t)}} \cdot \operatorname{clip}\left(\frac{\hat{\sigma}_i(t)}{0.10}, 0.50, 1.50\right).$$

The first factor controls the overall exploration scale, the square-root term gives larger bonuses to less-sampled arms, and the clipped standard-deviation multiplier increases caution for high-variance arms without allowing unstable early variance estimates to dominate.

Let $b(t) = \arg\max_i \hat{\mu}_i(t)$ be the current empirical best arm. Its score is

$$\mathrm{score}_{b(t)}(t) = \hat{\mu}_{b(t)}(t) + 0.95 B_{b(t)}(t).$$

For every other arm $i \neq b(t)$, VSCO keeps it as a challenger only if its optimistic estimate overlaps with the pessimistic estimate of the current empirical best arm:

$$\hat{\mu}_i(t) + B_i(t) \geq \hat{\mu}_{b(t)}(t) - B_{b(t)}(t).$$

If the condition holds, $\mathrm{score}_i(t) = \hat{\mu}_i(t) + B_i(t)$; otherwise, $\mathrm{score}_i(t) = -\infty$. The algorithm then selects

$$a_t = \arg\max_i \mathrm{score}_i(t),$$

pulls arm a_t, and updates $N_i(t)$, $S_i(t)$, and $Q_i(t)$.

Algorithm 1 Variance-Scaled Confidence-Overlap Strategy (VSCO)

Initialize $N_i, S_i, Q_i \leftarrow 0$ for all arms i
$m_T \leftarrow \min(4, \max(1, \lfloor T/(20K) \rfloor))$, $c_T \leftarrow 0.12T/(T+25)$, $w \leftarrow 0.95$
for $t = 0, \ldots, T-1$ **do**
  if $\exists i$ such that $N_i < m_T$ **then**
    Choose an arm a_t with $N_{a_t} < m_T$
  else
    Compute $\hat{\mu}_i = S_i/N_i$, $\hat{\sigma}_i = \sqrt{\max(0, Q_i/N_i - \hat{\mu}_i^2)}$ for all arms
    Compute $B_i = c_T \sqrt{\log(t+1)/N_i} \cdot \mathrm{clip}(\hat{\sigma}_i/0.10, 0.50, 1.50)$ for all arms
    $b \leftarrow \arg\max_i \hat{\mu}_i$
    Set $\mathrm{score}_b \leftarrow \hat{\mu}_b + wB_b$
    For $i \neq b$, set $\mathrm{score}_i \leftarrow \hat{\mu}_i + B_i$ if $\hat{\mu}_i + B_i \geq \hat{\mu}_b - B_b$, otherwise $-\infty$
    $a_t \leftarrow \arg\max_i \mathrm{score}_i$
  end if
  Pull a_t, observe r_t, and update $N_{a_t}, S_{a_t}, Q_{a_t}$
end for

3 Hyperparameters and Experiments

The final hyperparameters are summarized in Table 1. I selected these values by comparing performance on the public environments and additional stress environments.

Table 1: Final hyperparameters of VSCO.

Component	Final value	Reason
Warm-up pulls	$\min(4, \max(1, \lfloor T/(20K) \rfloor))$	Limits forced exploration
Confidence scale	$0.12 \cdot T/(T+25)$	Balances exploration and exploitation
Best-arm weight	0.95	Stabilizes current empirical best arm
Std. reference	0.10	Basic public noise scale
Std. multiplier clip	$[0.50, 1.50]$	Prevents extreme variance effects

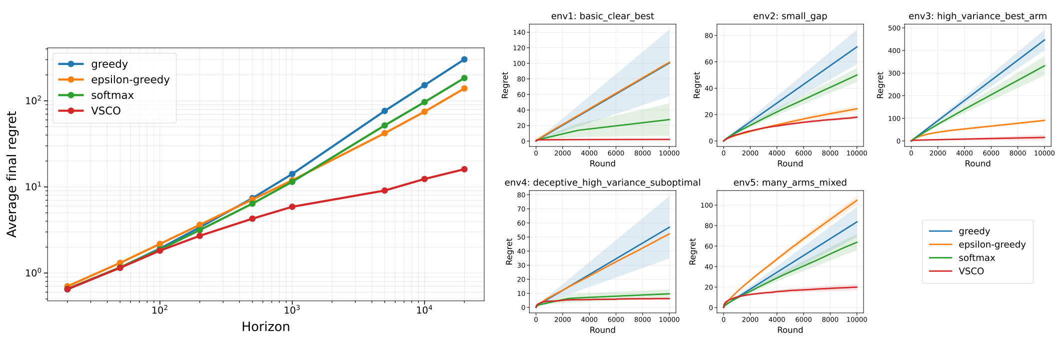
The most important coefficient was the confidence scale. I tested nearby values and found that smaller values such as 0.10 sometimes performed well on public seeds, but were less stable in longer-horizon and high-variance settings. A larger value gives more protection to high-variance best arms, but can also over-explore noisy suboptimal arms. I chose 0.12 because it gave strong public performance while remaining stable in additional stress tests. For example, the average final regrets for confidence bases $0.100, 0.105, 0.110, 0.115, 0.120$ on public $T = 10000$ tests were $13.720, 13.761, 16.731, 17.713, 13.257$, respectively.

Table 2 reports public results for $T = 10000$ and seeds $0, \ldots, 9$, where VSCO achieved the lowest regret in every environment.

Table 2: Public evaluation results with $T = 10000$ and seeds $0, \ldots, 9$. Lower regret is better.

Environment	VSCO regret	Lowest baseline	Baseline method
`env1_basic_clear_best`	**2.100**	100.610	Softmax
`env2_small_gap`	**15.485**	23.399	ϵ-greedy
`env3_high_variance_best_arm`	**2.698**	75.130	ϵ-greedy
`env4_deceptive_high_variance_suboptimal`	**4.096**	24.912	Softmax
`env5_many_arms_mixed`	**22.496**	40.036	Softmax

With 50 random seeds, the average regrets of Greedy, ϵ-greedy, Softmax, and VSCO were 151.781, 74.662, 96.636, and 12.365, respectively, showing that the improvement was not due to a few lucky seeds.



(a) Average regret over different horizons. (b) Regret curves on public environments at $T = 10000$.

Figure 1: Public regret results.

Figure 1 shows that VSCO remains competitive across horizons and that its regret usually grows mainly in the early stage before becoming almost flat. This supports the intended behavior: brief exploration, pruning of implausible arms, and concentration on the best arm.

4 Additional Tests and Discussion

To check robustness beyond the public environments, I created five stress environments testing small gaps, high-variance optimal arms, deceptive noisy suboptimal arms, many-arm settings, and very easy clear-best cases.

With $T = 10000$ and 50 seeds, the average final regrets of Greedy, ϵ-greedy, Softmax, and VSCO on these stress environments were 102.399, 91.847, 64.447, and **24.618**, respectively. VSCO performed best on average, although Greedy and Softmax were better in the easiest `stress_clear_best` case. This is expected because VSCO intentionally spends some samples on warm-up and confidence comparison to handle harder hidden cases.

VSCO works well because it does not explore all arms forever. Early on, many arms remain active because their confidence-style intervals are wide; later, arms whose optimistic estimates fall below the current best arm's pessimistic estimate are pruned, so most rounds are spent exploiting the best arm. Its limitations are late identification after unlucky high-variance samples and small warm-up cost in very easy cases. Nevertheless, VSCO uses sample mean, empirical variance, uncertainty estimation, and confidence-style comparison, and performed robustly across the tests.